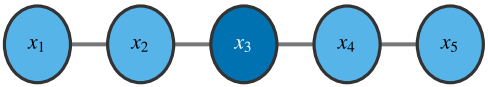
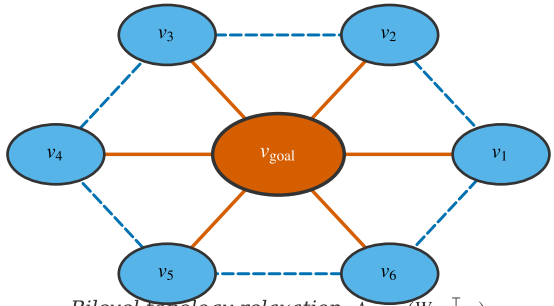
(a) Traditional 1D Lattice: Diameter $\Omega(n)$



Fixed nearest-neighbor kernel W_L, W_C, W_R

Receptive field expands as $O(T)$ steps

(b) Meta-Learned Graph NCA: Diameter ≤ 2 ($O(1)$ Receptive Field)



Bilevel topology relaxation: $A_{ij} = \sigma(W_\theta x_i^\top x_j)$